

CLASS 12 · CHEMISTRY

Coordination Compounds

Theory and worked examples

15 concepts

4 determinations

14 problem types

Recall sheet

How to use it. Part 1 builds the ideas in the order they depend on each other — read it straight through once. Part 2 works the four determinations that everything else is built from. Part 3 is every type of problem that gets set, each with what to recognise it by, the method, and a worked example. Part 4 is a single page to revise from once the rest is known.

The one rule worth carrying. Almost every question in this chapter reduces to comparing the crystal field splitting Δ_o with the pairing energy P . Larger Δ_o means pairing is cheaper than climbing, so the complex is low spin; smaller means the reverse. Shape, hybridisation, magnetism and colour all follow from that one comparison.

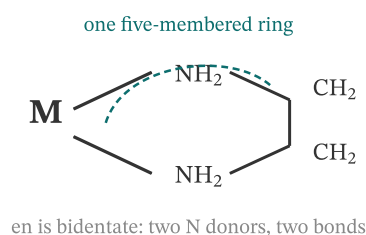
1 · The concepts, in the order they build

C1 Complex compound or double salt?

	Double salt	Complex compound
Behaviour in water	dissociates completely into all its simple ions	the coordination sphere stays intact ; only the ions outside it are released
Identity of the metal ion	retained, so it gives its usual tests	lost, so it does not give its usual tests
Example	Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

Hold on to this · This is the standard two-mark question, and it is always set the same way. FeSO_4 mixed with $(\text{NH}_4)_2\text{SO}_4$ in a 1:1 ratio gives Mohr's salt, a **double salt**, which dissociates fully and so **does** give the test for Fe^{2+} . CuSO_4 with aqueous ammonia in a 1:4 ratio gives the **complex** $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, in which Cu^{2+} is locked inside the coordination sphere and so gives **no** test for Cu^{2+} .

C2 Ligands, denticity and chelation



Chelate effect

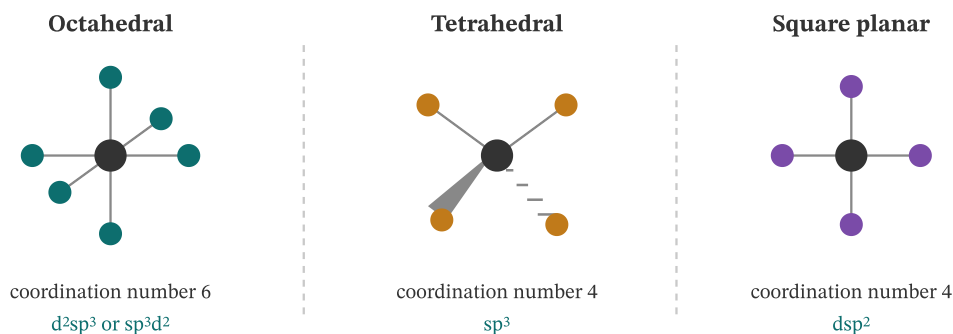
A ring-forming ligand holds far more tightly than the same number of separate ligands would. Five- and six-membered rings are the stable ones. en, ox^{2-} and EDTA^{4-} are the ones set.

Denticity	Donor sites	Examples
Monodentate	1	Cl^- , CN^- , NH_3 , H_2O , OH^- , NO_2^- , CO
Bidentate	2	ethane-1,2-diamine (en), oxalate ($\text{C}_2\text{O}_4^{2-}$), glycinate
Tridentate	3	diethylenetriamine (dien)
Tetradentate	4	triethylenetetramine (trien)
Pentadentate	5	EDTA acting through five sites
Hexadentate	6	EDTA⁴⁻ — its usual behaviour

- A ligand that binds through **two or more** sites forms a ring with the metal and is called a **chelating** ligand; the ring is a chelate. Chelation makes a complex markedly **more stable**.
- An **ambidentate** ligand can bind through *either* of two different atoms — NO_2^- through N (nitro) or through O (nitrito), and SCN^- through S (thiocyanato) or N (isothiocyanato). This is what gives rise to linkage isomerism (C7).

Hold on to this · **EDTA** is hexadentate and forms a very stable **octahedral** chelate. That is why it is used in the treatment of **lead poisoning** — it wraps around Pb^{2+} , and the resulting complex is soluble, unreactive and excreted by the kidneys. It is also used to estimate Ca^{2+} and Mg^{2+} in hard water.

C3 Oxidation number and coordination number



- **Coordination number (C.N.)** — the number of ligand **donor atoms** bonded to the central metal. Count donor *sites*, not ligand molecules: in $[\text{Co}(\text{en})_3]^{3+}$ there are three ligands but each ethane-1,2-diamine donates through **two** nitrogens, so the C.N. is **6**, not 3.
- **Oxidation number** — let it be x , add the charges of all the ligands, and set the total equal to the charge on the complex ion.

$$[\text{Co}(\text{en})_3]^{3+} : x + 3(0) = +3 \Rightarrow x = +3, \text{ C.N.} = 6$$

$$[\text{Fe}(\text{CN})_6]^{4-} : x + 6(-1) = -4 \Rightarrow x = +2, \text{ C.N.} = 6$$

$$[\text{Ni}(\text{CO})_4] : x + 4(0) = 0 \Rightarrow x = 0, \text{ C.N.} = 4$$

C4 IUPAC nomenclature — the rules in order

1. **Cation first, anion second**, as in any salt.
2. Within the complex, name the **ligands first, in alphabetical order**, then the metal. Alphabetise by the **ligand name**, ignoring the multiplying prefix.
3. Anionic ligands end in **-o**: chlorido, cyanido, hydroxido, oxalato, nitrito. Neutral ligands keep their names, except **aqua** (H_2O), **ammine** (NH_3), **carbonyl** (CO) and **nitrosyl** (NO).
4. Use di-, tri-, tetra- for simple ligands; use **bis-**, **tris-**, **tetrakis-** with brackets for ligands whose names already contain a multiplier, e.g. bis(ethane-1,2-diamine).
5. Give the **oxidation state of the metal in Roman numerals** in brackets, immediately after the metal, with no space.
6. If the complex is an **anion**, the metal takes the suffix **-ate** — and often its Latin stem: ferrate, cuprate, argentate, aurate, plumbate, stannate.

Formula	IUPAC name
$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Br}_3$	tetraamminediaquacobalt(III) bromide
$\text{K}_4[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(II)
$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$	tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II)
$[\text{CoCl}(\text{ONO})(\text{en})_2]^+$	chloridobis(ethane-1,2-diamine)(nitrito-O)cobalt(III) ion
$[\text{Ni}(\text{CO})_4]$	tetracarbonylnickel(0)

Common slip • Alphabetical order is by the ligand name, so **ammine comes before aqua** (m before q) — hence *tetraamminediaqua*, never the reverse. And in the fourth example, *chlorido* (c) precedes *ethane-1,2-diamine* (e), which precedes *nitrito* (n).

C5 Werner's theory and the AgNO₃ test

- Werner proposed two kinds of valency: **primary** (ionisable, satisfied by negative ions, shown outside the square brackets) and **secondary** (non-ionisable, satisfied by ligands, inside the brackets).
- Only the ions **outside** the coordination sphere are free in solution. So the number of moles of AgCl precipitated by excess AgNO₃ counts the chloride ions **outside** the brackets.

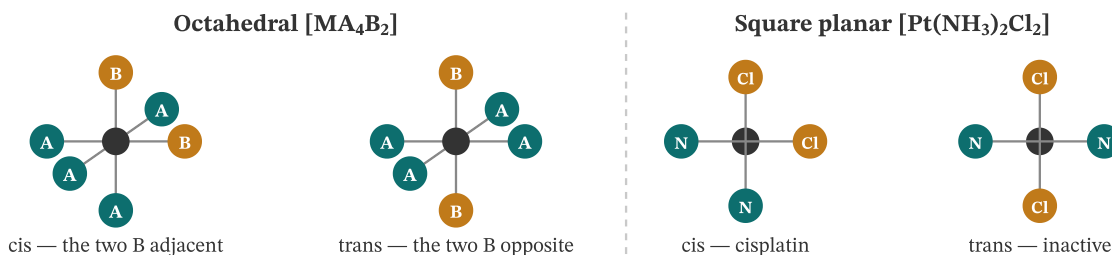
Complex	Ions in solution	mol AgCl per mol complex
[Co(NH ₃) ₆]Cl ₃	4 (1:3 electrolyte)	3
[Co(NH ₃) ₅ Cl]Cl ₂	3 (1:2 electrolyte)	2
[Co(NH ₃) ₄ Cl ₂]Cl	2 (1:1 electrolyte)	1
[Co(NH ₃) ₃ Cl ₃]	0 — a non-electrolyte	0

Hold on to this · More ions ⇒ higher molar conductivity. So the conductivity order is exactly the order of the table read upwards: [Co(NH₃)₃Cl₃] < [Co(NH₃)₄Cl₂]Cl < [Co(NH₃)₅Cl]Cl₂ < [Co(en)₃]Cl₃.

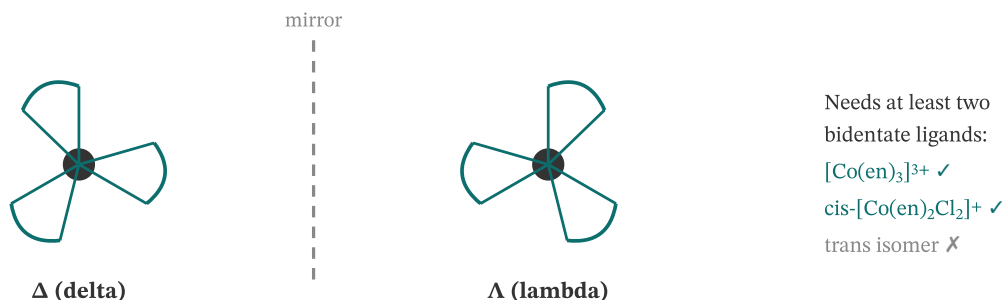
Careful with K₂[PtCl₄]: it ionises to 2K⁺ and [PtCl₄]²⁻. All four chlorides are inside the sphere, so it gives **no** precipitate with AgNO₃.

C6 Structural isomerism — the four kinds

Type	What differs	Example pair
Ionisation	which ion is inside the coordination sphere and which is the counter-ion	[Co(NH ₃) ₅ Cl]SO ₄ and [Co(NH ₃) ₅ SO ₄]Cl
Hydrate	whether water is a ligand or lattice water	[Cr(H ₂ O) ₆]Cl ₃ and [Cr(H ₂ O) ₅ Cl]Cl ₂ ·H ₂ O
Linkage	which atom of an ambidentate ligand is donating	[Co(NH ₃) ₅ (NO ₂)] ²⁺ (nitro, through N) and [Co(NH ₃) ₅ (ONO)] ²⁺ (nitrito, through O)
Coordination	how the ligands are shared between a complex cation and a complex anion	[Co(NH ₃) ₆][Cr(CN) ₆] and [Cr(NH ₃) ₆][Co(CN) ₆]

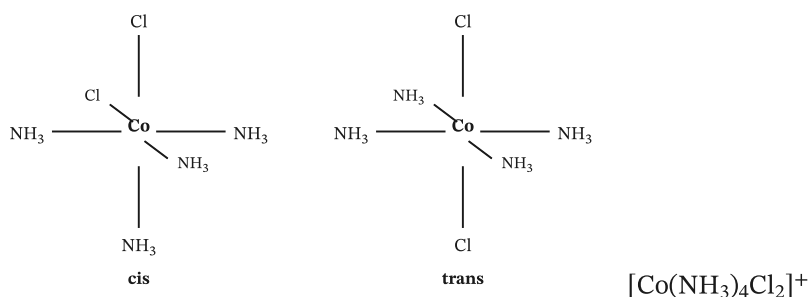


A tetrahedral MA_2B_2 shows none — all four positions are adjacent.



Non-superimposable mirror images — the two rotate plane polarised light equally and oppositely.

Geometrical (cis–trans) isomerism needs at least two different kinds of ligand, arranged so that two placements are genuinely distinct.



Shape and type	Geometrical isomerism?
Octahedral MA_4B_2	yes — cis and trans
Octahedral MA_3B_3	yes — facial (fac) and meridional (mer)
Octahedral $M(AA)_2B_2$, e.g. $[Co(en)_2Cl_2]^+$	yes — cis and trans
Octahedral MA_5B	no — only one arrangement exists
Square planar MA_2B_2 , e.g. $[Pt(NH_3)_2Cl_2]$	yes — cis and trans
Tetrahedral	no — all four positions are adjacent

- Optical isomerism** requires a non-superimposable mirror image. It is shown chiefly by octahedral complexes with **bidentate** ligands: $[Co(en)_3]^{3+}$ exists as a d and an l form, and $cis-[Co(en)_2Cl_2]^+$ is optically active while the *trans* form is not (it has a plane of symmetry).

Common slip · The one that is set as a “which does **not** show geometrical isomerism” distractor is the **MA_5B** type, such as $[Co(NH_3)_5NO_2]Cl_2$. With five identical ligands, every position for the sixth is equivalent.

C8 Valence bond theory — the procedure

Fe^{3+} is $3d^5$ — the ligand decides what happens next

Strong field — CN^-

3d  two 3d vacant

4s 4p 

d^2sp^3 — inner orbital, low spin

1 unpaired, $\mu = 1.73$ BM

Weak field — H_2O

3d  none vacant

4s 4p 4d 

sp^3d^2 — outer orbital, high spin

5 unpaired, $\mu = 5.92$ BM

- Find the oxidation state of the metal and write the d-electron count.
- Decide whether the ligands are **strong field** (they force pairing) or **weak field** (they do not) — see C10.
- If pairing occurs, inner (3d) orbitals become available and the complex uses d^2sp^3 — an **inner orbital** or **low spin** complex. If not, outer 4d orbitals are used and the hybridisation is sp^3d^2 — an **outer orbital** or **high spin** complex.
- Count the remaining unpaired electrons and get the magnetic moment.

Hybridisation	Shape	Coordination number
sp^3	tetrahedral	4
dsp^2	square planar	4
d^2sp^3	octahedral (inner orbital, low spin)	6
sp^3d^2	octahedral (outer orbital, high spin)	6

Hold on to this · Outer orbital complexes involve sp^3d^2 hybridisation and are high spin; inner orbital complexes are d^2sp^3 and low spin. That sentence is set verbatim as a fill-in-the-blanks.

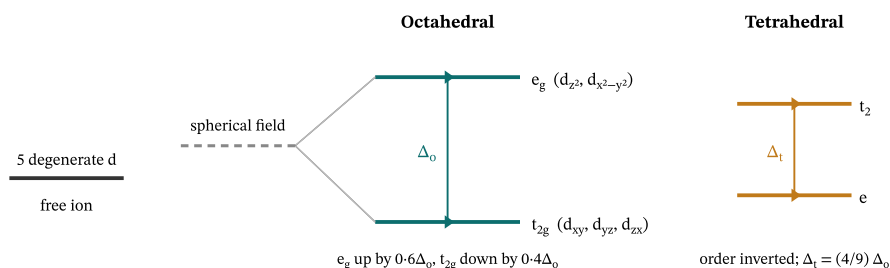
C9 Magnetic moment

$\mu = \sqrt{n(n+2)} \text{ BM}$ where n is the number of unpaired electrons

n	0	1	2	3	4	5
μ / BM	0	1.73	2.83	3.87	4.90	5.92

- $n = 0 \Rightarrow$ **diamagnetic**; $n \geq 1 \Rightarrow$ **paramagnetic**, and more strongly so as n rises.
- The question is often set backwards: given μ , find n, and hence deduce the geometry or the spin state.

C10 Crystal field theory and the spectrochemical series



- In an **octahedral** field the five d orbitals split into a lower t_{2g} set of three and an upper e_g set of two, separated by the crystal field splitting energy Δ_o .
- In a **tetrahedral** field the order is **inverted** — e below t_2 — and the splitting is much smaller, $\Delta_t = (4/9)\Delta_o$. Tetrahedral complexes are therefore almost always **high spin**.
- Whether electrons pair up depends on Δ_o against the pairing energy P .

The rule that answers every CFT question

$\Delta_o > P$ (strong field): electrons prefer to pair in t_{2g} → **low spin**, fewer unpaired electrons.

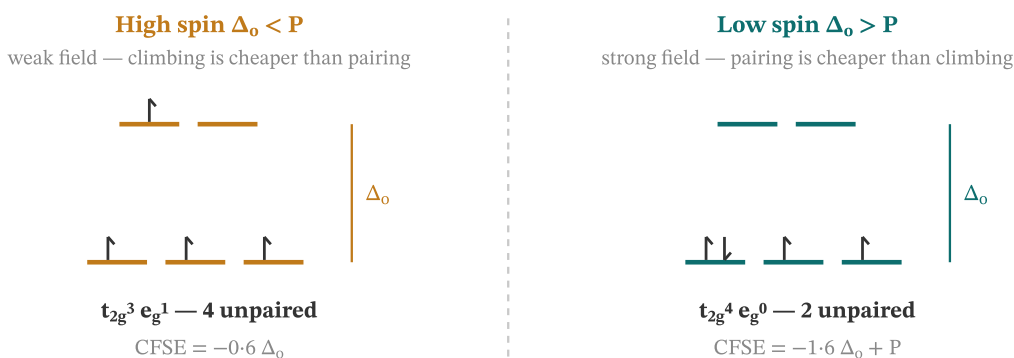
$\Delta_o < P$ (weak field): electrons prefer to occupy e_g → **high spin**, more unpaired electrons.

Spectrochemical series (weak field → strong field)



Memory hook • Only the two ends really need memorising: the **halides and water are weak field** (high spin, often paramagnetic), while **CN[−] and CO are strong field** (low spin, often diamagnetic). NH_3 sits in the middle and is treated as strong field for cobalt(III) and nickel(II).

C11 The d⁴ case, which is the one always set



Condition	Configuration	Unpaired electrons	Spin state
$\Delta_o > P$ (strong field)	$t_{2g}^4 e_g^0$	2	low spin
$\Delta_o < P$ (weak field)	$t_{2g}^3 e_g^1$	4	high spin

The same reasoning gives the d^5 , d^6 and d^7 cases. d^6 is worth knowing separately: strong field gives $t_{2g}^6 e_g^0$, **zero** unpaired electrons and a diamagnetic complex — which is why $[Fe(CN)_6]^{4-}$ and $[Co(NH_3)_6]^{3+}$ are both diamagnetic.

C12 Colour — the d-d transition

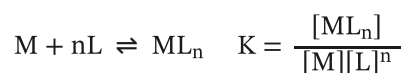
A transition metal complex is coloured because Δ_o happens to correspond to the energy of **visible light**. An electron absorbs a photon and is promoted from t_{2g} to e_g — a **d-d transition** — and the colour seen is the **complement** of the colour absorbed.

- A larger Δ_o means a higher absorbed frequency, so changing the ligand changes the colour. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ is pale blue; $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is deep blue.
- A complex with **no d electrons** (d^0 , e.g. Sc^{3+} , Ti^{4+}) or a **full d subshell** (d^{10} , e.g. Zn^{2+} , Cu^+) has no possible d-d transition and is therefore **colourless**.

Hold on to this · The two-part answer that earns full marks: **the d subshell is partially filled, so a d-d transition is possible**. Both halves are needed — Say *partially filled*, not *unpaired electrons*: $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ are low-spin d^6 , so they have **no** unpaired electrons at all, and both are still coloured.

C13 Stability of complexes

Stability is measured by the **stability constant K** for



- **A larger K means a more stable complex.**
- Stability rises with the **charge on the metal ion**, with a **smaller ionic radius**, and strongly with **chelation** — which is why EDTA complexes are so stable.

C14 Two complexes worth knowing individually

Complex	Oxidation state	Hybridisation	Shape	Magnetism
$[\text{Ni}(\text{CO})_4]$	0	sp^3	tetrahedral	diamagnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	+2	dsp^2	square planar	diamagnetic
$[\text{NiCl}_4]^{2-}$	+2	sp^3	tetrahedral	paramagnetic (2 unpaired)
$[\text{Fe}(\text{CN})_6]^{4-}$	+2	d^2sp^3	octahedral	diamagnetic

Hold on to this · The $[\text{Ni}(\text{CN})_4]^{2-}$ against $[\text{NiCl}_4]^{2-}$ pair is the single most-set comparison in this unit. Same metal, same oxidation state, same coordination number — yet different shapes and different magnetism, entirely because CN^- is a strong field ligand that forces the d^8 electrons to pair, while Cl^- is weak field and does not.

C15 Importance and uses

- **Qualitative analysis** — Fe^{3+} with SCN^- gives a blood-red complex; Cu^{2+} with excess NH_3 gives the deep blue $[\text{Cu}(\text{NH}_3)_4]^{2+}$; Ni^{2+} with dimethylglyoxime gives a red precipitate.
- **Extraction of metals** — silver and gold are leached as their cyanide complexes, $[\text{Ag}(\text{CN})_2]^-$.
- **Purification** — nickel by Mond's process, through the volatile $[\text{Ni}(\text{CO})_4]$.
- **Biological systems** — haemoglobin (Fe), chlorophyll (Mg), vitamin B_{12} (Co), carboxypeptidase (Zn).
- **Medicine** — *cis*-platin, *cis*- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, in cancer chemotherapy; EDTA in lead poisoning.

2 · The four standard determinations

D1 Oxidation number, coordination number and the number of ions

Worked on $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$:

1. **Coordination number** — count donor atoms inside the brackets: five NH_3 plus one Cl = **6**.
2. **Charge on the complex ion** — two chlorides sit outside, so the ion must be $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$.
3. **Oxidation number** — $x + 5(0) + (-1) = +2$, so $x = +3$.
4. **Ions in solution** — one complex cation and two chlorides = **3 ions**, a 1:2 electrolyte, giving **2 mol AgCl** with excess AgNO_3 .

■ determined

D2 Valence bond treatment of $[\text{Fe}(\text{CN})_6]^{4-}$

1. $x + 6(-1) = -4$, so Fe is **+2**; Fe^{2+} is **$3d^6$** .
2. CN^- is a **strong field** ligand, so it forces the six d electrons to pair up in the three t_{2g} -type orbitals.
3. Two 3d orbitals are thereby vacated, so the metal hybridises as **d^2sp^3** — an **inner orbital** complex.
4. Shape **octahedral**; unpaired electrons **$n = 0$** ; $\mu = 0$, so the complex is **diamagnetic**.

Contrast $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$: water is weak field, so the d^6 configuration stays high spin with $n = 4$, giving sp^3d^2 hybridisation, an outer orbital complex and $\mu = 4.90 \text{ BM}$.

■ determined

D3 Crystal field treatment of a d^4 ion

1. Place the four electrons in the split levels, following the rule in C10.

$\Delta_o > P$: $t_{2g}^4 e_g^0 \Rightarrow$ two electrons paired, **$n = 2$** , low spin, $\mu = 2.83 \text{ BM}$

$\Delta_o < P$: $t_{2g}^3 e_g^1 \Rightarrow$ all four unpaired, **$n = 4$** , high spin, $\mu = 4.90 \text{ BM}$

2. Read the answer off: the strong-field complex is the less paramagnetic of the two, and the two are **different colours**, since the larger Δ_o absorbs at a higher frequency. Note that Δ_o fixes *which* colour is absorbed, not how intense the colour is — so do not write that the strong-field complex is ‘more deeply coloured’.

■ determined

D4 Deducing a molecular formula from conductivity data

“CoBr₃·4NH₃·2H₂O has a molar conductivity corresponding to a 1:3 electrolyte. Write its molecular formula and IUPAC name.”

1. A **1:3 electrolyte** gives one cation and three anions, so **four ions** in all.
2. Three of these must be the three Br[−], so **all three bromides lie outside** the coordination sphere and the complex ion carries a charge of **+3**.
3. Everything else — four NH₃ and two H₂O — must therefore be **inside**, giving a coordination number of 6, which is what cobalt(III) prefers.



Naming it: ligands alphabetically, ammine before aqua, then the metal with its oxidation state, then the counter-ion —

tetraamminediaquacobalt(III) bromide

■ **determined**

3 · Every type of problem that is set

T1 Oxidation number and coordination number

RECOGNISE IT BY

“What is the oxidation number of the metal / the coordination number?”

METHOD

Follow D1. Count **donor atoms**, not ligand molecules — a bidentate ligand contributes 2 and EDTA contributes 6.

WORKED EXAMPLE

For $[\text{Cr}(\text{en})_2(\text{H}_2\text{O})\text{Cl}]^{2+}$, find the oxidation number of chromium and its coordination number.

1. donor atoms: $2 \text{ en} \times 2 = 4$, plus 1 H_2O , plus 1 $\text{Cl} = 6$
2. $x + 2(0) + 0 + (-1) = +2$

Cr is +3; coordination number 6

T2 Write the IUPAC name from the formula

RECOGNISE IT BY

A formula is given; the name is wanted.

METHOD

Apply C4 in order: ligands alphabetically (by ligand name, ignoring prefixes), anionic ligands in -o, oxidation state in Roman numerals, -ate if the complex is an anion.

WORKED EXAMPLE

Name (a) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$ and (b) $[\text{CoCl}(\text{ONO})(\text{en})_2]^+$.

- (a) The cation is $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$, so Pt is +4; the anion is $[\text{PtCl}_4]^{2-}$, so Pt is +2. Name: **tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II)**.
- (b) $x + (-1) + (-1) + 0 = +1$, so Co is +3. Alphabetical order is chlorido, ethane-1,2-diamine, nitrito. Name: **chloridobis(ethane-1,2-diamine)(nitrito-O)cobalt(III) ion**.

T3 Write the formula from the name

RECOGNISE IT BY

The reverse direction, or “deduce the formula from the conductivity”.

METHOD

Work out the charge on the complex ion from the metal's oxidation state and the ligand charges, then add counterions to balance. For a conductivity question, follow D4.

WORKED EXAMPLE

$\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ has a molar conductivity corresponding to a 1:3 electrolyte. Write the molecular formula and the IUPAC name.

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Br}_3$ — tetraamminediaquacobalt(III) bromide

Fully worked in D4.

T4 Identify the type of isomerism

RECOGNISE IT BY

A pair of formulae, or a single complex, with “state the type of isomerism”.

METHOD

Ask what has moved. Across the brackets $\Rightarrow$ **ionisation** (or hydrate, if it is water). A different donor atom of the same ligand $\Rightarrow$ **linkage**. Ligands swapped between two complex ions $\Rightarrow$ **coordination**. Nothing moved, only the arrangement in space $\Rightarrow$ **geometrical** or **optical**.

WORKED EXAMPLE

State the type of isomerism shown by (a) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and (b) $[\text{Co}(\text{en})_3]^{3+}$.

- (a) **Ionisation isomerism** — its isomer is $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$, in which the sulphate is the ligand and chloride the counter-ion. The two give different precipitation tests: the first gives a white precipitate with BaCl_2 , the second with AgNO_3 .
- (b) **Optical isomerism** — three bidentate ligands about an octahedral centre give a propeller-shaped ion with no plane of symmetry, so d and l forms exist.

T5 Draw the geometrical isomers

RECOGNISE IT BY

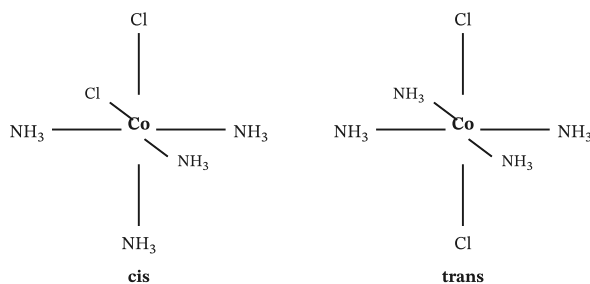
“Draw the geometrical isomers of ...”

METHOD

First check the type is capable of it (C7). Draw the octahedron as a square plane with two axial positions, then put the two odd ligands **adjacent** (cis) and **opposite** (trans).

WORKED EXAMPLE

Draw the geometrical isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.



Neither isomer is optically active here: the trans form has a plane of symmetry, and so does the cis form of this particular complex.

T6 Which complex does not show geometrical isomerism?**RECOGNISE IT BY**

A list of four complexes, one of which is the exception.

METHOD

Classify each by type against the table in C7. The answer is almost always the **MA₅B** member of the list, or a tetrahedral one.

WORKED EXAMPLE

Which of the following will **not** show geometrical isomerism? (a) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

Answer · (c). It is of the MA₅B type — five identical ammine ligands and one nitro — so every possible position for the odd ligand is equivalent and only one structure exists. (a) and (b) are MA₄B₂ types and (d) is square planar MA₂B₂; all three give cis and trans forms.

T7 Valence bond theory: hybridisation, shape and magnetism**RECOGNISE IT BY**

“State the hybridisation, shape and magnetic behaviour of ...”

METHOD

Follow D2: oxidation state → d-count → strong or weak field → pairing → hybridisation → n → μ. Give **all** the parts the question asks for; each carries a mark.

WORKED EXAMPLE

For $[\text{Fe}(\text{CN})_6]^{4-}$: (a) the oxidation number of iron, (b) the number of unpaired electrons, (c) the hybridisation and magnetic behaviour.

- (a) $x + 6(-1) = -4$, so **Fe is +2**, and Fe^{2+} is $3d^6$.
- (b) CN^- is strong field, so all six d electrons pair in t_{2g} : **n = 0**.
- (c) **d^2sp^3** , octahedral, inner orbital, and **diamagnetic** ($\mu = 0$).

T8 Why two complexes of the same metal differ in shape**RECOGNISE IT BY**

The $[\text{Ni}(\text{CN})_4]^{2-}$ against $[\text{NiCl}_4]^{2-}$ comparison, or its equivalents.

METHOD

Both have Ni^{2+} , d^8 , coordination number 4. The **ligand field strength** alone decides: a strong field ligand forces pairing, freeing a 3d orbital for dsp^2 and square planar geometry; a weak field ligand does not, leaving sp^3 and tetrahedral.

WORKED EXAMPLE

The geometry of $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ are _____ and _____ respectively.

Complex	Ligand	Hybridisation	Geometry	Magnetism
$[\text{Ni}(\text{CN})_4]^{2-}$	strong field	dsp^2	square planar	diamagnetic
$[\text{NiCl}_4]^{2-}$	weak field	sp^3	tetrahedral	paramagnetic, n = 2

T9 Crystal field configuration

RECOGNISE IT BY

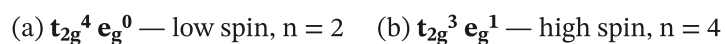
“On the basis of CFT, write the electronic configuration of a d^4 ion if $\Delta_o > P$ and if $\Delta_o < P$.”

METHOD

Use the one rule from C10 and fill the levels. Then, if asked, count unpaired electrons and give μ .

WORKED EXAMPLE

Write the CFT electronic configuration of a d^4 ion when (a) $\Delta_o > P$ and (b) $\Delta_o < P$.



T10 Conductivity order and number of ions

RECOGNISE IT BY

“Arrange the following complexes in increasing order of conductivity in solution.”

METHOD

Count the ions each formula releases — that is, count what lies **outside** the square brackets. More ions means higher conductivity; a complex with nothing outside the brackets is a **non-electrolyte** and conducts least.

WORKED EXAMPLE

Arrange in increasing order of conductivity: $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Co}(\text{en})_3]\text{Cl}_3$, $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

1. ions released: 0, 4, 3, 2 respectively

Increasing conductivity



T11 Precipitate with silver nitrate

RECOGNISE IT BY

“Will it give a precipitate with AgNO_3 ? How does it ionise?”

METHOD

Only **ionisable** halide — outside the brackets — precipitates. Count it, and say how many moles of AgCl or AgBr result per mole of complex.

WORKED EXAMPLE

How does $\text{K}_2[\text{PtCl}_4]$ ionise in water? Will it give a precipitate with AgNO_3 ?



Answer · No precipitate. All four chlorides are inside the coordination sphere, held by non-ionisable secondary valencies, so there are **no free Cl^- ions** in solution for the silver ion to react with.

The related assertion–reason question is worth watching: $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ indeed gives no precipitate, but the reason “it contains neutral ligands only” is **false** — the three chlorides are anionic ligands. Assertion true, reason false.

T12 Complex or double salt?

RECOGNISE IT BY

Two mixtures in stated molar ratios, one giving the metal ion test and one not.

METHOD

A **1:1** ratio of two simple salts usually means a **double salt**, which dissociates completely and gives the normal tests. A ratio matching a coordination number (1:4, 1:6) means a **complex**, in which the metal ion is not free.

WORKED EXAMPLE

FeSO₄ mixed with (NH₄)₂SO₄ in a 1:1 molar ratio gives the test for Fe²⁺, but CuSO₄ with aqueous ammonia in a 1:4 molar ratio does not give the test for Cu²⁺. Explain.

Answer · The first gives **Mohr's salt**, FeSO₄·(NH₄)₂SO₄·6H₂O, a **double salt**. In water it dissociates completely into Fe²⁺, NH₄⁺ and SO₄²⁻, so free Fe²⁺ is present and gives its usual test. The second gives the **complex** [Cu(NH₃)₄]SO₄, in which Cu²⁺ is bound inside the coordination sphere by non-ionisable secondary valencies. It *does* ionise — into [Cu(NH₃)₄]²⁺ and SO₄²⁻ — but the **complex ion itself does not dissociate**, so no free Cu²⁺ is available and the test fails.

T13 EDTA and chelation

RECOGNISE IT BY

A passage about EDTA in medicine or water treatment, then short parts.

METHOD

Four facts answer nearly every version: EDTA is **hexadentate**; it forms a **very stable octahedral chelate**; in lead poisoning it **sequesters Pb²⁺** into a soluble complex that is excreted; and in water analysis it is used to **estimate Ca²⁺ and Mg²⁺**.

WORKED EXAMPLE

(i) What type of ligand is EDTA? (ii) Why is EDTA used in the treatment of lead poisoning? (iii) What is the geometry of the EDTA complex?

- (i) A **hexadentate chelating** ligand — it donates through two nitrogen atoms and four carboxylate oxygens.
- (ii) It **chelates the Pb²⁺ ion**, forming a complex of very high stability constant. The lead is thereby removed from the tissues, and the water-soluble complex is excreted through the kidneys.
- (iii) **Octahedral**, since the coordination number is 6.

T14 Explaining colour or magnetism**RECOGNISE IT BY**

“Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ coloured but $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ colourless?”

METHOD

Both answers rest on the **d-electron count**. Colour requires unpaired electrons in the d subshell and a possible **d–d transition**; a d^0 or d^{10} ion has neither. Magnetism requires unpaired electrons, so count n and use $\mu = \sqrt{n(n+2)}$.

WORKED EXAMPLE

Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ violet while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless?

Answer · Ti^{3+} is **3d¹**. Its single d electron can absorb visible light and be promoted from t_{2g} to e_g — a **d–d transition** — and the complex appears in the complementary colour. Sc^{3+} is **3d⁰**: with no d electron at all, no d–d transition is possible, so it absorbs nothing in the visible region and is colourless. Zn^{2+} (d^{10}) is colourless for the opposite reason — the subshell is full, so there is no vacancy to be promoted into.

4 · Coordination compounds recall sheet

Hybridisation $\Rightarrow$ shape

$sp^3 \rightarrow$ tetrahedral
 $dsp^2 \rightarrow$ square planar
 $d^2sp^3 \rightarrow$ octahedral, **inner**, low spin
 $sp^3d^2 \rightarrow$ octahedral, **outer**, high spin

Magnetic moment

$\mu = \sqrt{n(n+2)}$ BM
 $n = 0, 1, 2, 3, 4, 5 \rightarrow 0, 1.73, 2.83, 3.87, 4.90, 5.92$
 $n = 0$ diamagnetic; $n \geq 1$ paramagnetic

CFT in one line

Octahedral: t_{2g} below e_g
Tetrahedral: **inverted**, $\Delta_t = (4/9)\Delta_o$
 $\Delta_o > P \rightarrow$ low spin; $\Delta_o < P \rightarrow$ high spin
 d^4 : $t_{2g}^4 (n=2)$ or $t_{2g}^3 e_g^1 (n=4)$

Field strength

Weak: I^- , Br^- , Cl^- , F^- , H_2O
Strong: NH_3 , en , NO_2^- , CN^- , CO

The four Ni / Fe complexes

$[Ni(CO)_4] \rightarrow 0, sp^3$, tetrahedral, diamagnetic
 $[Ni(CN)_4]^{2-} \rightarrow dsp^2$, square planar, diamagnetic
 $[NiCl_4]^{2-} \rightarrow sp^3$, tetrahedral, paramagnetic
 $[Fe(CN)_6]^{4-} \rightarrow d^2sp^3$, diamagnetic

Ions and conductivity

$[M(NH_3)_6]Cl_3 \rightarrow 4$ ions, 3 AgCl
 $[M(NH_3)_5Cl]Cl_2 \rightarrow 3$ ions, 2 AgCl
 $[M(NH_3)_4Cl_2]Cl \rightarrow 2$ ions, 1 AgCl
 $[M(NH_3)_3Cl_3] \rightarrow 0$ ions, no ppt.

Isomerism, four structural kinds

Ionisation — ion swaps across the brackets
Hydrate — water inside or as lattice water
Linkage — NO_2^- / SCN^- donor atom
Coordination — ligands swapped between two complex ions

Geometrical isomerism

Yes: MA_4B_2 , MA_3B_3 , $M(AA)_2B_2$, square planar MA_2B_2
No: MA_5B , and all tetrahedral
Optical: $[M(en)_3]$, cis- $[M(en)_2Cl_2]$

Nomenclature traps

Alphabetise by **ligand name**, ignore prefixes
ammine before aqua
bis/tris for en, not di/tri
anionic complex $\rightarrow$ metal takes **-ate**

Colour

Due to **unpaired d electrons** and a **d-d transition**
 d^0 and d^{10} are **colourless**

Double salt vs complex

Double salt (Mohr's) — dissociates fully, **gives** the metal test
Complex ($[Cu(NH_3)_4]SO_4$) — sphere intact, **no** metal test

EDTA

Hexadentate chelating ligand, octahedral
Lead poisoning; estimation of Ca^{2+}/Mg^{2+}